

Provincial Agricultural Drought Mitigation Planning: Optimal LDRRMF Allocation for Cavite Province

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1 Introduction

In municipal climate-adaptation planning, local governments are faced with the challenge of deploying scarce fiscal capital to mitigate severe environmental hazards. In agricultural economies such as the Province of Cavite, prolonged moisture deficits triggered by the El Niño Southern Oscillation (ENSO) threaten crop yields, lower soil water retention, and jeopardize food security. Historically, local disaster resource allocations have relied heavily on uncalibrated intuition or uniform ad-hoc distributions, often leaving the most ecologically vulnerable crop regions severely underfunded.

To move toward an audit-ready, evidence-based governance paradigm, operational research integrates predictive machine learning with constrained mathematical programming. By combining multi-year satellite climatological data with statutory fiscal boundaries, local disaster councils can formulate resource distribution as a continuous **Linear Programming Problem (LPP)**. The objective is to deploy public resources optimally to maximize risk-weighted disaster protection while guaranteeing legal compliance, municipal operational readiness, and regional agricultural equity.

2 Statement of the Problem

The Provincial Disaster Risk Reduction and Management Council (PDRRMC) of Cavite is preparing its fiscal allocation plan for the peak dry season (February to May). The province comprises twenty-three (23) local government units ($n = 23$), consisting of 16 municipalities and 7 component cities.

A two-stage analytical pipeline has been developed:

1. A Random Forest regression model trained on multi-year NASA POWER agro-meteorological data predicted the mean dry-season root-zone soil wetness ($\overline{\text{GWETROOT}}_i \in [0, 1]$) for each LGU. The empirical drought risk score is defined as the complementary moisture deficit:

$$R_i = 1 - \overline{\text{GWETROOT}}_i, \quad \forall i \in \{1, 2, \dots, 23\}. \quad (1)$$

2. A linear programming model that distributes the pre-disaster preparedness budget (x_i) to maximize total risk-weighted impact subject to statutory public finance laws and physical framework constraints.

The optimization model follows two types of rules: **national legal mandates** (from RA 10121) and **project benchmark parameters** (realistic example values chosen for this case study).

1. **Legal Minimum for Local Calamity Funds (Statutory Law):** Under **Republic Act No. 10121, Section 21, Paragraph 1**, every local government unit (LGU) must set aside at least five percent (5%) of its regular yearly revenue for the Local Disaster Risk Reduction and Management Fund (LDRRMF, denoted as T).

2. **The 30% Quick Response Fund Reserve (Statutory Law):** Under **RA 10121, Section 21, Paragraph 2** and **NDRRMC–DBM–DILG Joint Memorandum Circular No. 2013-1, Section 5.1**, exactly thirty percent (30%) of the LDRRMF must be set aside as a Quick Response Fund (QRF, denoted as $Q = 0.30T$). By law, this emergency money is on stand-by for relief work and cannot be budgeted or spent before an official State of Calamity is declared.

3. **The 70% Pre-Disaster Preparedness Budget (Statutory Law):** Because 30% is locked for emergency relief, the remaining seventy percent (70%) of the fund is used for proactive disaster prevention and preparedness ($B = 0.70T$). The linear programming model allocates only this 70% preparedness budget:

$$\sum_{i=1}^{23} x_i \leq B. \quad (2)$$

4. **Total Provincial Budget Example (Benchmark Value):** While the 5% rule is set by law, the exact peso amount depends on the province’s revenue. For this research project, we use a representative benchmark of:

$$T = \text{PHP } 50,000,000.$$

Following the legal 70/30 split, this sets the preparedness budget ceiling at $B = \text{PHP } 35,000,000$, and leaves $Q = \text{PHP } 15,000,000$ in the locked emergency reserve.

5. **Minimum Baseline Allocation (Benchmark Value):** The law does not specify an exact peso floor for each town. To ensure every municipality receives enough money to maintain weather sensors, transport, and local staff, we set a minimum policy floor of $L_i = \text{PHP } 400,000$:

$$x_i \geq 400,000 \quad \forall i \in \{1, \dots, 23\}. \quad (3)$$

6. **Maximum Allocation Cap (Benchmark Value):** To prevent any single city or town from taking too much of the shared budget, we set an administrative cap of fifteen percent (15%) of the preparedness budget ($U_i = 0.15B = \text{PHP } 5,250,000$):

$$x_i \leq 5,250,000 \quad \forall i \in \{1, \dots, 23\}. \quad (4)$$

7. **Upland Farming Cluster Share (Benchmark Value):** According to the *Cavite Provincial Development and Physical Framework Plan (PDPFP)*, the province’s coffee, cacao, and rain-fed crops are located in the nine (9) upland towns:

$$\mathcal{U} = \{\text{Alfonso, Amadeo, Gen. Emilio Aguinaldo, Indang, Magallanes, Maragondon, Mendez, Silang, Tagaytay City}\}$$

Because rain-fed farms face higher drought risk than irrigated lowland areas, we set a policy rule that these 9 towns must collectively receive at least sixty percent (60%) of the total preparedness budget ($0.60B = \text{PHP } 21,000,000$). The remaining balance (up to 40%, or $\text{PHP } 14,000,000$) is available for the 14 lowland towns and cities:

$$\sum_{i \in \mathcal{U}} x_i \geq 0.60B. \quad (5)$$

3 Tasks

As quantitative policy consultants to the Provincial Government of Cavite, you are tasked with completing the following deliverables:

1. **Mathematical Formulation:** Formally define the continuous Linear Programming Problem. Clearly label decision variables, the risk-weighted objective function, and express all policy constraints in standard mathematical inequality notation. Show how the 30% Quick Response Fund (Q) is explicitly partitioned under RA 10121, Section 21.

1. Statutory Fund Partition and Parameter Definitions

In strict adherence to Section 21 of Republic Act No. 10121 and Joint Memorandum Circular No. 2013-1, the benchmark Local Disaster Risk Reduction and Management Fund ($T = \text{PHP } 50,000,000$) is partitioned into two statutory components prior to optimization:

$$Q = 0.30 T = \text{PHP } 15,000,000 \quad (\text{Quick Response Fund, stand-by relief reserve}) \quad (6)$$

$$B = 0.70 T = \text{PHP } 35,000,000 \quad (\text{Statutory Pre-Disaster Preparedness Budget}) \quad (7)$$

Because Q is legally encumbered for post-disaster response during a declared State of Calamity, only the preparedness ceiling B is available for proactive allocation.

Let the operational parameters and regional topology of Cavite be defined as follows:

- $\mathcal{M} = \{1, 2, \dots, 23\}$: The index set denoting all 23 local government units (16 municipalities and 7 component cities) of Cavite ($|\mathcal{M}| = 23$).
- $\mathcal{U} \subset \mathcal{M}$: The subset containing the nine (9) designated upland agricultural LGUs ($|\mathcal{U}| = 9$), leaving 14 lowland LGUs in $\mathcal{M} \setminus \mathcal{U}$.
- $d_i \in \{0, 1\}$: A deterministic parameter encoding regional agricultural zoning:

$$d_i = \begin{cases} 1, & \text{if } i \in \mathcal{U} \\ 0, & \text{if } i \in \mathcal{M} \setminus \mathcal{U} \end{cases}$$

- $R_i \in [0, 1]$: The empirical drought risk score of LGU i , defined by the complement of predicted root-zone soil wetness: $R_i = 1 - \overline{\text{GWETROOT}}_i$.
- $B = \text{PHP } 35,000,000$: The total provincial statutory preparedness budget ceiling.
- $L_i = \text{PHP } 400,000$: The mandatory operational survival floor for each LGU $i \in \mathcal{M}$.
- $U_i = 0.15 B = \text{PHP } 5,250,000$: The individual equity absorption ceiling for each LGU $i \in \mathcal{M}$.

2. Primal Problem Formulation

The resource distribution model is formulated as the following continuous Linear Programming Problem (LPP):

$$\begin{aligned} \text{maximize} \quad & Z = \sum_{i \in \mathcal{M}} R_i x_i \\ \text{subject to:} \quad & \sum_{i \in \mathcal{M}} x_i \leq B \quad (\text{Statutory Preparedness Budget Ceiling}) \\ & \sum_{i \in \mathcal{M}} d_i x_i \geq 0.60 B \quad (\text{Zonal Upland Agricultural Guarantee}) \quad (8) \\ & x_i \geq L_i, \quad \forall i \in \mathcal{M} \quad (\text{Operational Survival Floor}) \\ & x_i \leq U_i, \quad \forall i \in \mathcal{M} \quad (\text{Equity Absorption Ceiling}) \\ & x_i \geq 0, \quad \forall i \in \mathcal{M} \quad (\text{Non-Negativity}) \end{aligned}$$

where each decision variable $x_i \in \mathbb{R}_{\geq 0}$ denotes the continuous fiscal appropriation granted to municipality or city i .

3. Reduction to Canonical Standard Form

To transform the primal formulation into the canonical standard form ($\min \mathbf{c}^T \mathbf{x}$ subject to $\mathbf{Ax} = \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$), we perform the following transformations:

- (a) Convert maximization to minimization by negating the objective coefficients: $\max Z \equiv -\min(-Z)$.
- (b) Introduce slack variable $s_1 \geq 0$ for the statutory budget ceiling constraint.
- (c) Introduce surplus variable $e_1 \geq 0$ for the zonal upland guarantee constraint.
- (d) Introduce surplus variables $v_i \geq 0$ for each $i \in \mathcal{M}$ for the operational survival floor constraints.
- (e) Introduce slack variables $u_i \geq 0$ for each $i \in \mathcal{M}$ for the equity absorption ceiling constraints.

The scalar equality system is expressed as:

$$\begin{aligned}
& \text{minimize} && \sum_{i \in \mathcal{M}} -R_i x_i \\
& \text{subject to:} && \sum_{i \in \mathcal{M}} x_i + s_1 = 35,000,000 \\
& && \sum_{i \in \mathcal{M}} d_i x_i - e_1 = 21,000,000 \\
& && x_i - v_i = 400,000, \quad \forall i \in \mathcal{M} \\
& && x_i + u_i = 5,250,000, \quad \forall i \in \mathcal{M} \\
& && x_i, v_i, u_i \geq 0 \quad (\forall i \in \mathcal{M}), \quad s_1, e_1 \geq 0
\end{aligned} \tag{9}$$

4. Matrix Representation

The system comprises $m = 48$ linear equality constraints and $n = 71$ continuous non-negative variables. The augmented solution vector $\mathbf{x} \in \mathbb{R}^{71}$, cost vector $\mathbf{c} \in \mathbb{R}^{71}$, and requirement vector $\mathbf{b} \in \mathbb{R}^{48}$ are partitioned in block form as:

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_{23 \times 1} \\ s_1 \\ e_1 \\ \mathbf{v}_{23 \times 1} \\ \mathbf{u}_{23 \times 1} \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} -R_1 \\ \vdots \\ -R_{23} \\ 0 \\ 0 \\ \mathbf{0}_{23 \times 1} \\ \mathbf{0}_{23 \times 1} \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 35,000,000 \\ 21,000,000 \\ 400,000 \cdot \mathbf{1}_{23 \times 1} \\ 5,250,000 \cdot \mathbf{1}_{23 \times 1} \end{bmatrix}$$

The structural constraint matrix $\mathbf{A} \in \mathbb{R}^{48 \times 71}$ is assembled as:

$$\mathbf{A} = \begin{bmatrix} \mathbf{1}_{23 \times 1}^T & 1 & 0 & \mathbf{0}_{23 \times 1}^T & \mathbf{0}_{23 \times 1}^T \\ \mathbf{d}_{23 \times 1}^T & 0 & -1 & \mathbf{0}_{23 \times 1}^T & \mathbf{0}_{23 \times 1}^T \\ \mathbf{I}_{23} & \mathbf{0}_{23 \times 1} & \mathbf{0}_{23 \times 1} & -\mathbf{I}_{23} & \mathbf{0}_{23 \times 23} \\ \mathbf{I}_{23} & \mathbf{0}_{23 \times 1} & \mathbf{0}_{23 \times 1} & \mathbf{0}_{23 \times 23} & \mathbf{I}_{23} \end{bmatrix} \tag{10}$$

where $\mathbf{I}_{23}$ is the 23×23 identity matrix, $\mathbf{1}_{23 \times 1}$ is a column vector of ones, and $\mathbf{d} = [d_1, \dots, d_{23}]^T$ is the static upland indicator vector. Because every decision, slack, and surplus variable belongs to $\mathbb{R}_{\geq 0}$, the problem defines a bounded convex polyhedron in $\mathbb{R}^{71}$ solvable via standard continuous Primal Simplex algorithms.

2. **Algorithmic Implementation Formulation:** To execute the optimization numerically via modern solvers such as `scipy.optimize.linprog` (using `method='highs'`), the continuous LPP is cast into the canonical bounded upper-inequality form:

$$\begin{aligned} & \text{minimize} && \mathbf{c}^T \mathbf{x} \\ & \text{subject to} && \mathbf{A}_{\text{ub}} \mathbf{x} \leq \mathbf{b}_{\text{ub}} \\ & && \boldsymbol{\ell} \leq \mathbf{x} \leq \mathbf{u} \end{aligned} \tag{11}$$

Rather than augmenting the system with 46 auxiliary slack and surplus variables for individual municipal floors and ceilings, modern implementations handle box constraints directly via variable bounds. This reduces the structural inequality matrix to only $m = 2$ global coupling constraints over $n = 23$ decision variables:

- **Objective Vector** ($\mathbf{c} \in \mathbb{R}^{23}$): Since `linprog` minimizes by default, the risk maximization objective is converted by setting:

$$\mathbf{c} = -\mathbf{R} = -[R_1, R_2, \dots, R_{23}]^T \tag{12}$$

- **Decoupled Box Bounds** ($\boldsymbol{\ell}, \mathbf{u} \in \mathbb{R}^{23}$): The survival floors and absorption ceilings are isolated from the row constraints and supplied via the `bounds` parameter:

$$\ell_i = L_i = \text{PHP } 400,000, \quad u_i = U_i = \text{PHP } 5,250,000, \quad \forall i \in \mathcal{M} \tag{13}$$

- **Coupling Inequality System** ($\mathbf{A}_{\text{ub}} \in \mathbb{R}^{2 \times 23}$, $\mathbf{b}_{\text{ub}} \in \mathbb{R}^2$): The remaining macro-policy constraints consist of the global statutory budget ceiling and the zonal upland guarantee. Inverting the greater-than inequality of the upland guarantee to conform to the standard $\leq$ format yields:

$$\mathbf{A}_{\text{ub}} = \begin{bmatrix} \mathbf{1}_{23}^T \\ -\mathbf{d}^T \end{bmatrix} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ -d_1 & -d_2 & \cdots & -d_{23} \end{bmatrix} \in \mathbb{R}^{2 \times 23} \tag{14}$$

and the corresponding right-hand side capacity vector:

$$\mathbf{b}_{\text{ub}} = \begin{bmatrix} B \\ -0.60 B \end{bmatrix} = \begin{bmatrix} 35,000,000 \\ -21,000,000 \end{bmatrix} \in \mathbb{R}^2 \tag{15}$$

This compact formulation matches the mathematical inputs passed directly to the solver routine while eliminating redundant basis inversions.

3. **Policy Solution Analysis:** Analyze the resulting optimal allocation. Which local government units hit the 15% ceiling (U_i), and which remain at the PHP 400,000 baseline floor (L_i)? Explain the resource distribution logic: how does the optimization model balance giving baseline survival funds to every town, prioritizing high-risk upland areas, and preventing over-concentration of public funds?

1. Allocation Summary Across LGUs

- **Ceiling LGUs** ($x_i^* = U_i = \text{PHP } 5,250,000$): Five upland municipalities reach the 15% upper bound: **Amadeo, General Emilio Aguinaldo, Indang, Magallanes, and Mendez**. Together, they receive PHP 26,250,000 (75.00% of the budget).

- **Marginal LGU** ($L_i < x_i^* < U_i$): **Alfonso** receives **PHP 1,950,000** (5.57% of the budget), which exhausts the remaining balance of the preparedness fund.
- **Floor LGUs** ($x_i^* = L_i = \text{PHP } 400,000$): The remaining 17 LGUs (14 lowland localities and three upland towns: Silang, Tagaytay City, and Maragondon) stay at the operational floor of **PHP 400,000** each, totaling PHP 6,800,000 (19.43% of the budget).

2. Distribution Mechanics

The Linear Programming solution follows directly from the interaction between the objective coefficients and the constraints:

- Baseline Operational Reserves:** The model first assigns the lower bound ($L_i = \text{PHP } 400,000$) to all 23 LGUs, committing PHP 9,200,000 to ensure minimum administrative readiness across the province.
- Marginal Allocation by Risk Rank:** The remaining discretionary budget (PHP 25,800,000) is assigned by the Simplex algorithm to the municipalities with the highest objective coefficients ($R_i = 0.550000$).
- Upper Bound Truncation:** The individual ceiling ($U_i = \text{PHP } 5,250,000$) limits each town to an additional PHP 4,850,000 above the floor. Five of the tied upland LGUs absorb PHP 24,250,000 ($5 \times \text{PHP } 4,850,000$), leaving a residual balance of PHP 1,550,000. This remaining amount is assigned to Alfonso, bringing its total to PHP 1,950,000.
- Zonal Constraint Check:** Total funding directed to upland municipalities equals PHP 29,400,000 (84.00%), satisfying the $\geq 60\%$ (PHP 21,000,000) provincial target without making that constraint strictly binding.

4. Scenario Analysis: Climate Stress-Testing under an Intensifying El Niño Advisory

Grounded in the official DOST-PAGASA Climate Advisory (Press Release No. 225) warning of an intensifying “Very Strong” El Niño event, conduct a counterfactual sensitivity experiment to evaluate the resilience of the combined machine learning and linear programming allocation pipeline. Rather than manually altering budgetary quotas, apply an empirical climatological perturbation directly to the dry-season evaluation features (February–May):

$$\text{PRECTOTCORR}' = 0.70 \times \text{PRECTOTCORR}, \quad \text{T2M}' = \text{T2M} + 1.5^\circ\text{C} \quad (16)$$

representing a 30% reduction in precipitation (consistent with PAGASA’s official dry spell / below-normal criteria) coupled with a $+1.5^\circ\text{C}$ surface air temperature anomaly.